

RUDRA KHAITAN

+91-8700914076 | ✉ rudrakhaitan79@gmail.com |  [Rudra Khaitan](#) |  [Rukh79](#)

EDUCATION

Birla Institute of Technology and Science Pilani, (BITS Pilani)

October 2022 - July 2027

Dual Major, MSc. Biological Sciences, B.E. Computer Science

Pilani, India

◦ GPA: 8.58/10.00; Biology Major GPA: 9.77/10.00

EXPERIENCE

NVIDIA Corporation - Performance Design-Verification Intern

May 2026 - July 2026

- Worked on CPU performance & verification workflows, for benchmark to micro-benchmark conversion
- Developed a tool for classification of benchmarks based on a 26D vector created from PMU-stats

Structure-function mapping of genomic regions using Deep Learning & Graph Theory

Nov 2025 - May 2026

PI: Dr. Shibasish Chowdhury | Department of Biological Sciences, BITS Pilani

- Developed a multimodal framework combining sequence embeddings from DNA-BERT2, 14 biophysical DNA shape features (deepDNASHape), & graph neural networks for human promoter prediction
- Built pipelines for processing 59k promoter and GC-matched sequences from hg38 and EPDnew
- Demonstrated that modeling DNA structural topology improves promoter prediction performance, achieving AUROC 0.89 on held-out test data

Bhabha Atomic Research Centre, Govt. of India (BARC), Visakhapatnam, India

May 2024 - July 2024

PI: Dr. S. Pradhan | Atomic and Molecular Physics Division, BARC

- Simulated Fresnel-Kirchhoff diffraction in Python to validate a custom optical microscope
- Designed validation scripts to verify simulation outputs, improving result reliability & reproducibility

PROJECTS

LLM Fine-tuning: Intelligent Tutoring System (ITS) that follows Learning By Teaching

July 2025 - May 2026

PI: Dr. Jagat Sesh Challa | Department of Computer Science and Information Systems, BITS Pilani

- Fine-tuned LLMs under Learn-by-Teaching framework for structured knowledge reinforcement
- Applied Bloom's taxonomy to generate labeled question-answer datasets from standard textbooks

Transformer-Based Flood Mapping from Bi-Temporal SAR Imagery

Jan 2026 - May 2026

- Implemented ChangeFormer (PVT-v2-B2, All-MLP decoder) & a Swin-UNet hybrid in PyTorch over 10,255 Sentinel-1 triplets, adapting ImageNet patch-embed weights to 1- & 2-channel SAR input
- Addressed 34:1 class imbalance with a combined Focal and Dice objective, reaching mIoU 0.945 and mDice 0.971 on a held-out 1,539-tile split

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, Matplotlib, PyTorch, scikit-learn), C/C++

Deep Learning: Vision transformers (Swin, PVT, SegFormer), graph neural networks (GATv2, PyTorch Geometric), semantic and change-detection segmentation, mixed-precision training, transfer learning across modality mismatch

Bioinformatics: DNA-BERT-2, deepDNASHape, hg38, EPDnew

Tools: Git, Linux/Ubuntu, Jupyter Notebook, Google Colab, Perforce

Experimental Biology: Molecular cloning, spectrophotometry (NanoDrop), gel electrophoresis

HONORS AND AWARDS

Merit Cum Need Scholarship, top 3% of students for academic excellence | BITS Pilani Oct 2022 - Present

TEACHING EXPERIENCE

Lead Teaching Assistant, Introduction to Bioinformatics, BITS Pilani Digital

Aug 2025 - Present

- Lead TA for Introduction to Bioinformatics; TA for General Biology and Object-Oriented Programming
- Developed & graded assessments, resolved student queries, & supported delivery of the courses